



# Sound

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**By Sanskriti**

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# Introduction to Sound

- **Sound** is a form of energy that produces the sensation of hearing in our ears.
  - **Speed of Light:**  $\sim 3 \times 10^8$  m/s
  - **Speed of Sound in air:**  $\sim 344$  m/s
- Light travels almost instantaneously, while sound takes time to reach us.

# How is Sound Produced?

- Sound is produced by **vibrations**.
- All vibrating bodies produce sound.
- Vibrations are sometimes visible (e.g., musical instruments).
- In some cases, they can only be **felt** or **heard**.

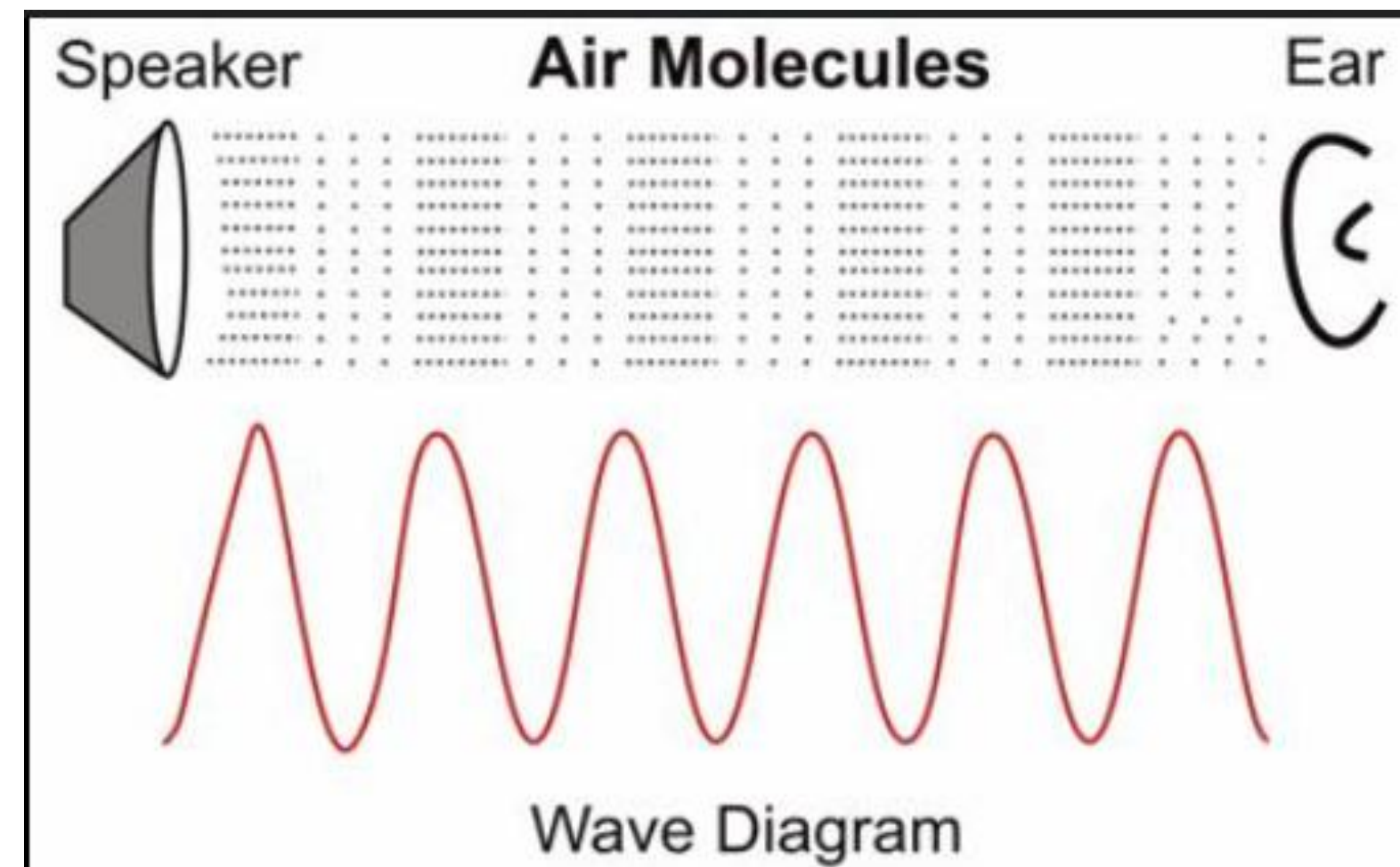
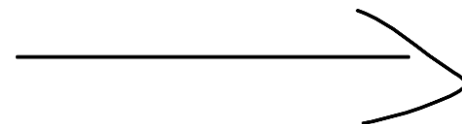
# How does Sound Propagate?

- Vibrations cause molecules in air near the mouth to disturb.
- These molecules **vibrate to and fro** and disturb neighboring molecules.
- This continues till molecules near the listener's ear **start vibrating**.
- Vibrating molecules **cause vibrations in the eardrum** → Sound is heard.

longitudinal



transverse



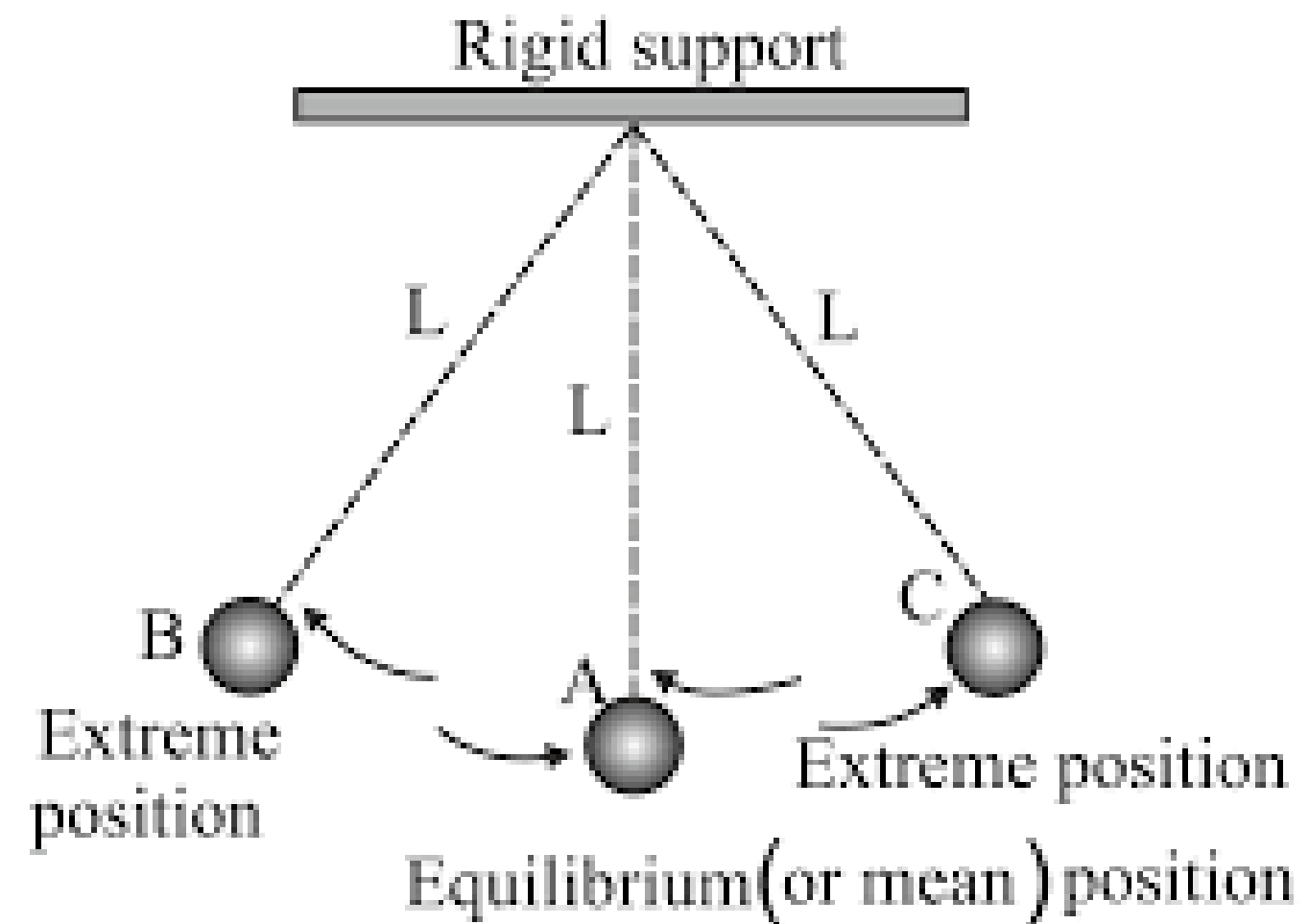


# Vibrations or Oscillations

- A **vibrating body** produces sound.
- **To-and-fro motion** of a body about its mean position = **Vibration/Oscillation**
  - **Slow vibrations:** Low sound
  - **Fast vibrations:** High sound
- **Oscillatory motion** = Swing-like motion (e.g., pendulum)

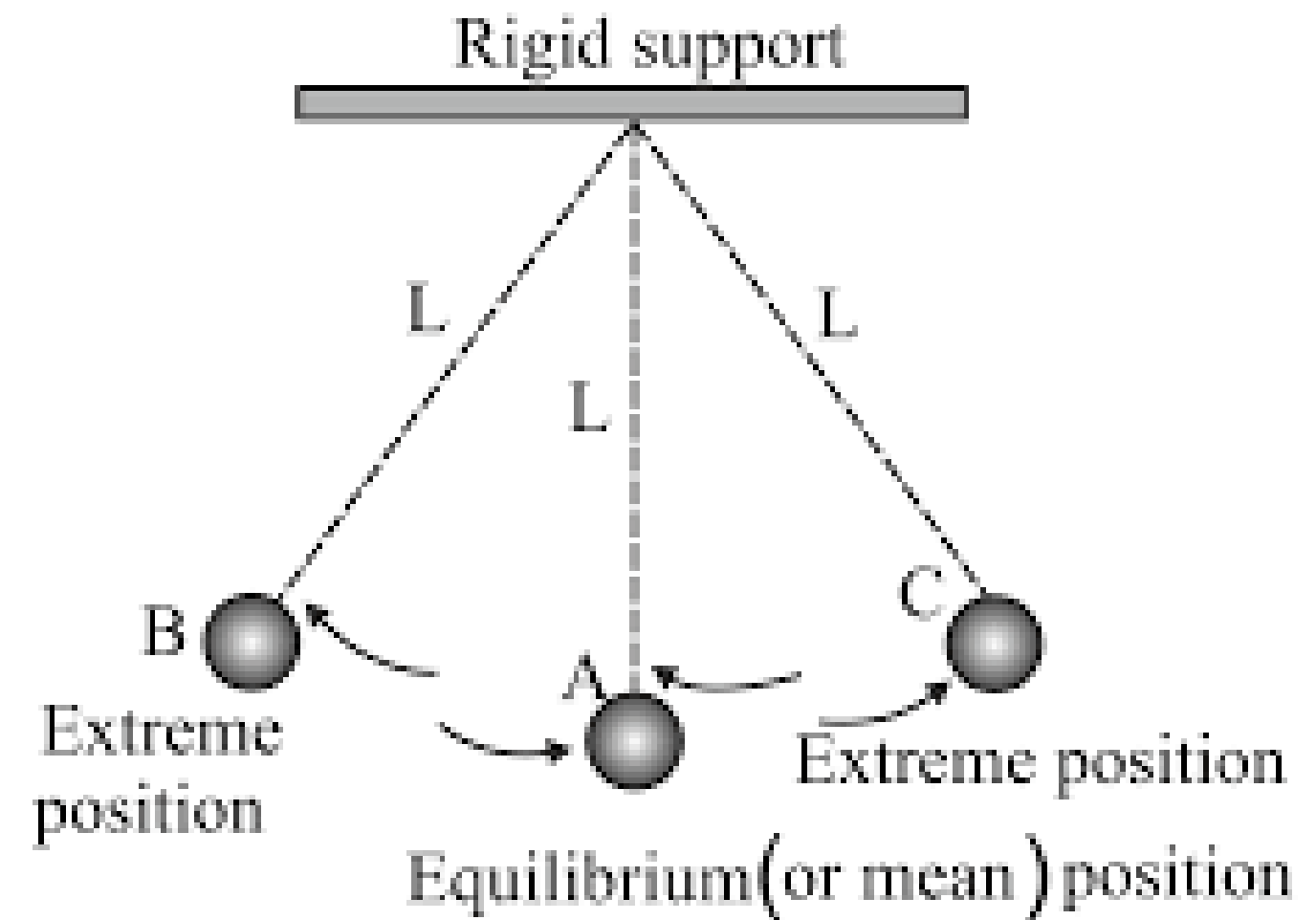
# Examples of Oscillatory Motion

- Pendulum
- Vibration of a stretched string
- Movement of vocal cords
- Movement of diaphragm in loudspeakers



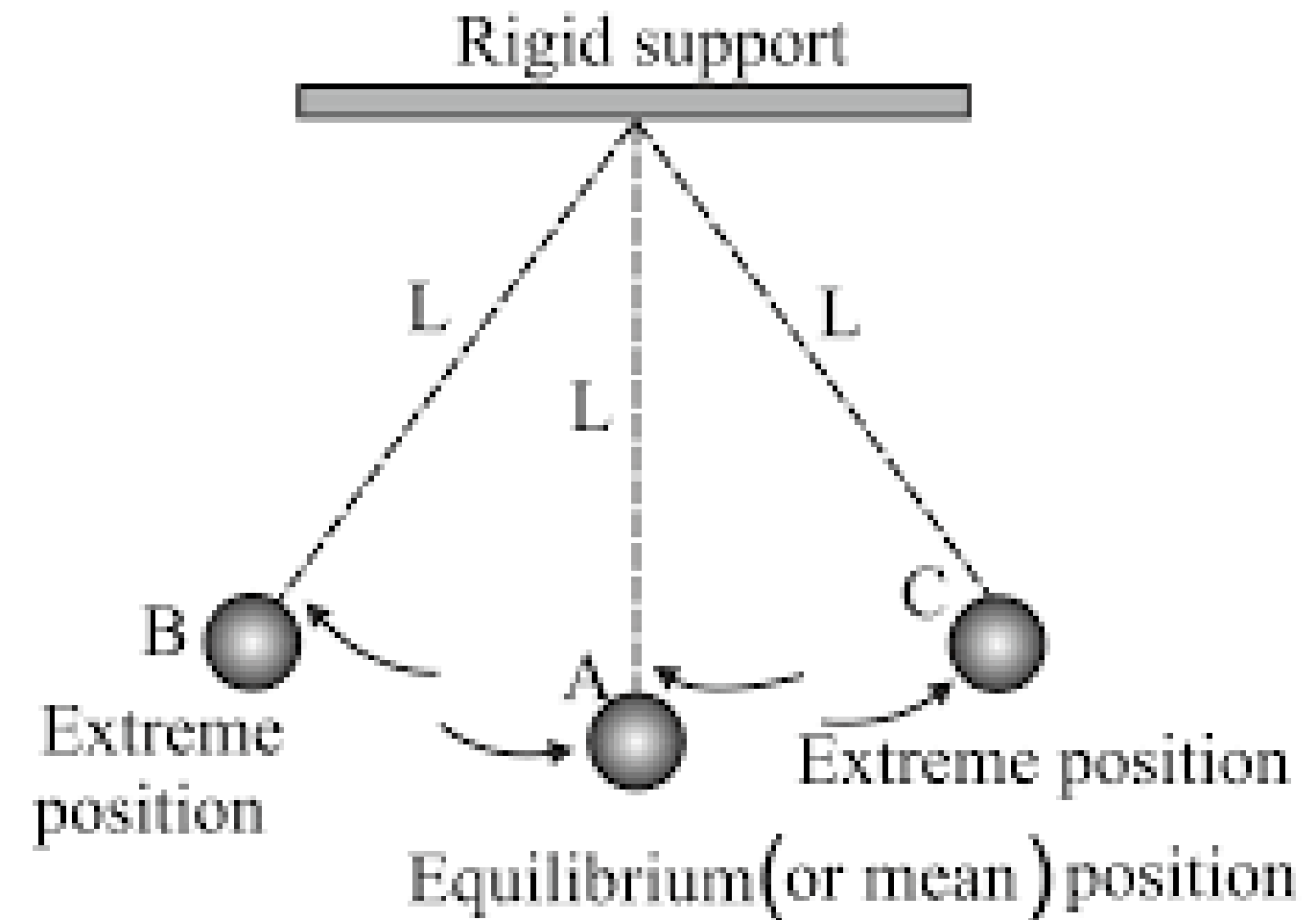
# Amplitude of Oscillation (A)

- **Definition:** Maximum displacement from the mean position.
- Shown by **AC** or **AB** in pendulum diagram.
- Larger amplitude = louder sound.
- Amplitude **gradually decreases** due to air resistance.



# Time Period (T)

- Time taken to complete **one oscillation**.
- Denoted by **T**, measured in seconds.
- Small amplitude = faster oscillation.





# Frequency of Oscillation

- Number of oscillations per second = **Frequency**

- Formula:

Frequency ( $\nu$ ) =  $1/\text{Time Period (T)}$

- **SI Unit:** Hertz (Hz)

1 Hz = 1 oscillation/second

# Audible and Inaudible Sound

- **Audible range:** 20 Hz to 20,000 Hz
  - Below 20 Hz: **Infrasonic**
  - Above 20,000 Hz: **Ultrasonic**
- Dogs, bats, dolphins hear **ultrasonic sounds**.

# Ultrasound and Its Uses

- Used to:**

- Drive away rats and cockroaches
- Detect cracks in metal
- Clean parts of machines
- Locate icebergs, submarines (SONAR)

# Sound Needs a Medium to Travel

- Sound needs a **material medium** to travel.
- It cannot travel in a **vacuum**.

## Speed of Sound

Depends on the **medium**:

- **Air:** 330 m/s
- **Water:** 1500 m/s
- **Steel:** 6000 m/s

**Formula:**

$$\text{Speed of Sound} = \frac{\text{Distance}}{\text{Time}}$$

# Reflection of Sound & Echo

- Sound reflects from **hard surfaces**.
- If reflected sound reaches after 0.1 seconds → **Echo**.

**Echo formula:**

$$\text{Distance} = \frac{1}{2} \times \text{Speed} \times \text{Time}$$

A ship hears echo after 1s, speed = 1500 m/s, What is depth of the sea?



# Characteristics of Sound

Sound is characterized by:

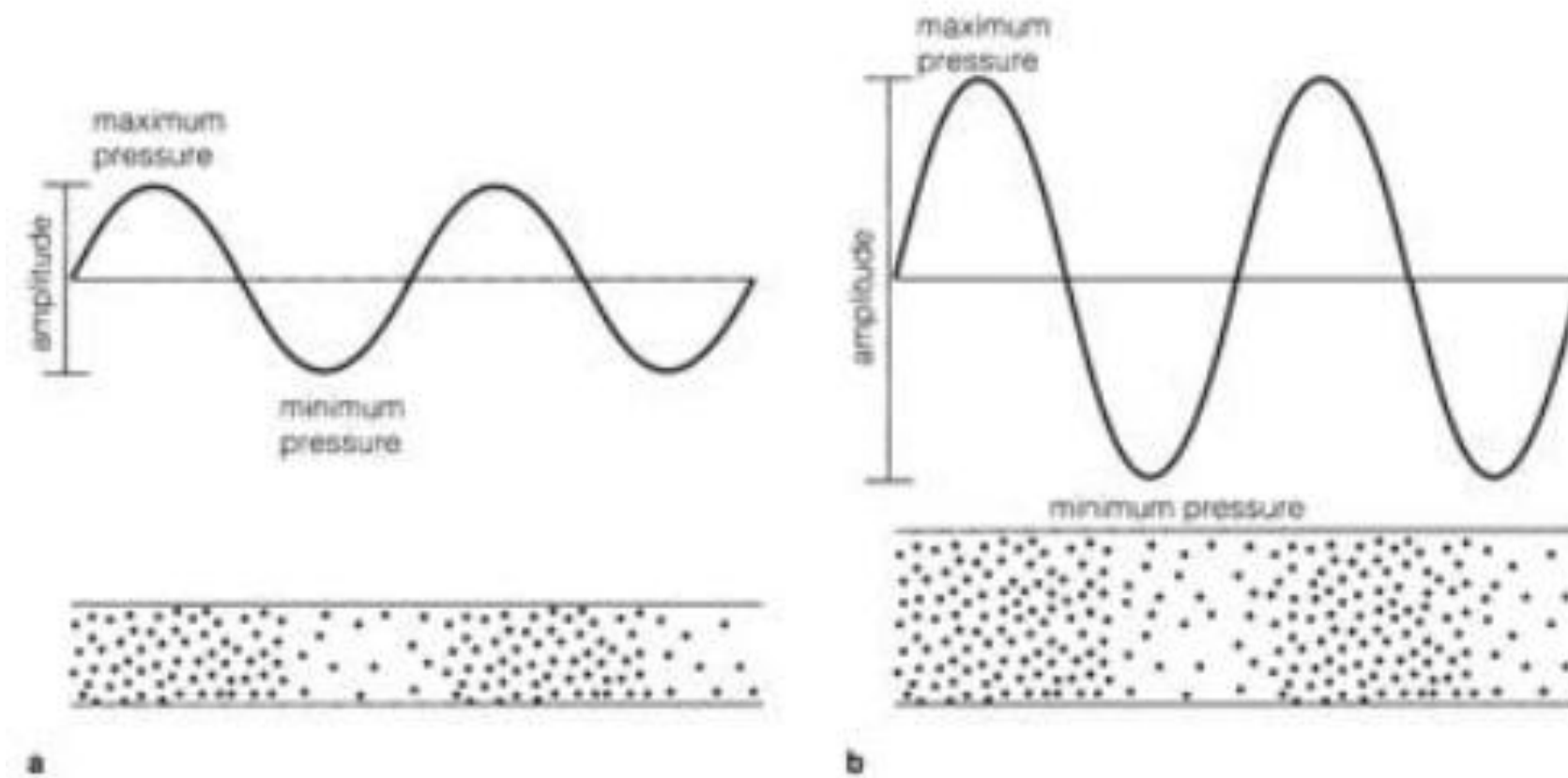
**1.Loudness**

**2.Pitch**

**3.Quality or Timbre**

# Loudness

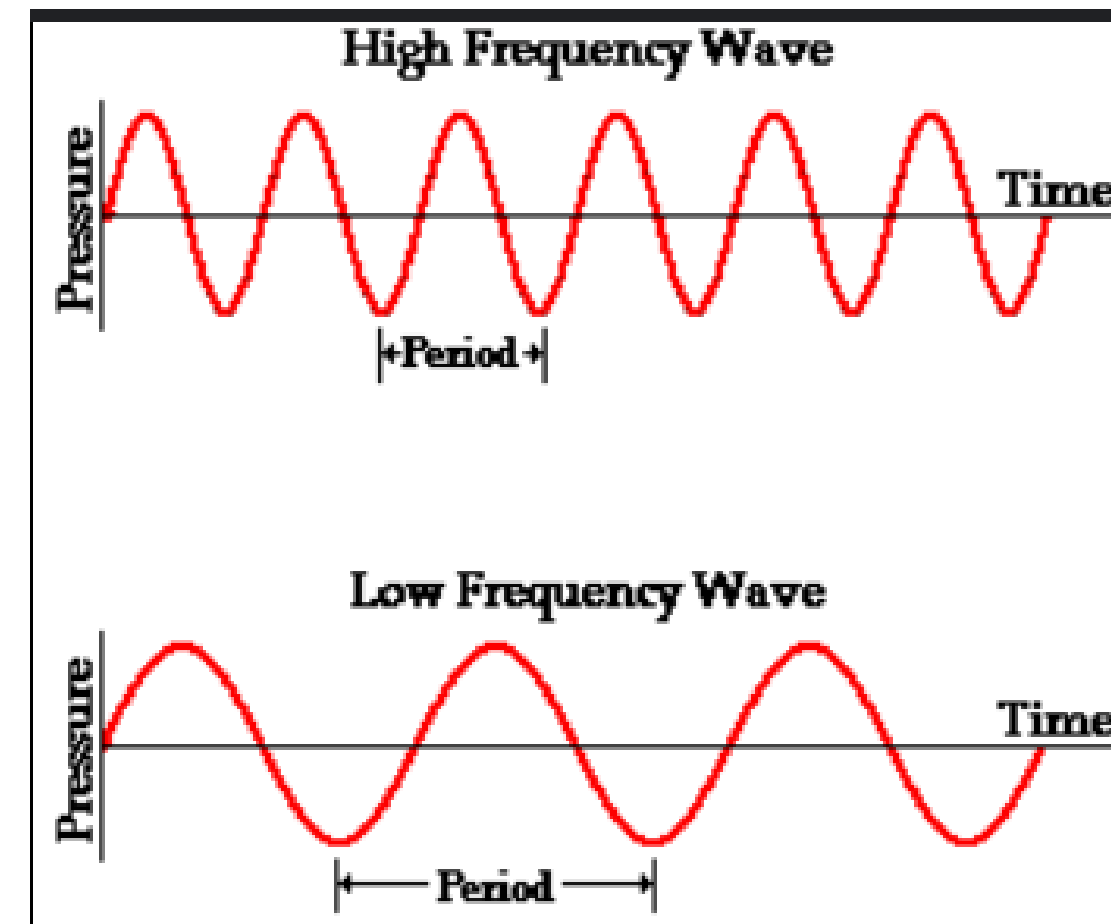
- Depends on **amplitude** of vibration.
- Greater amplitude = **louder sound**.
- Also depends on the **amount of air** that vibrates.
- Measured in **decibels (dB)**.

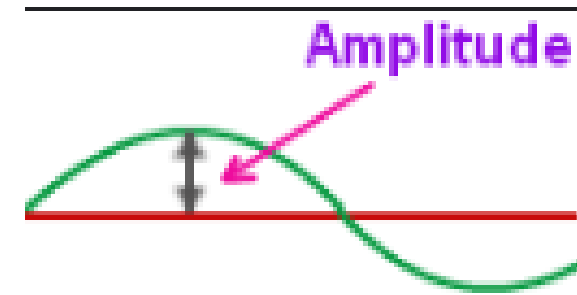




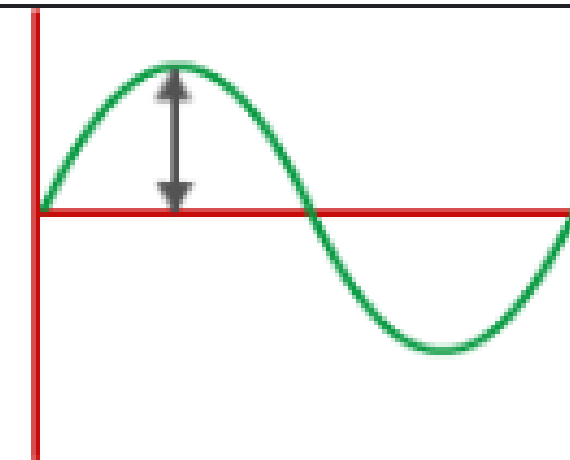
# Pitch

- Pitch = **shrillness** of sound.
- Depends on **frequency**:
- High frequency = High pitch
- Low frequency = Low pitch
- Voice of a child or woman → **higher pitch**
- **Stretching a membrane** tightly = increases pitch.

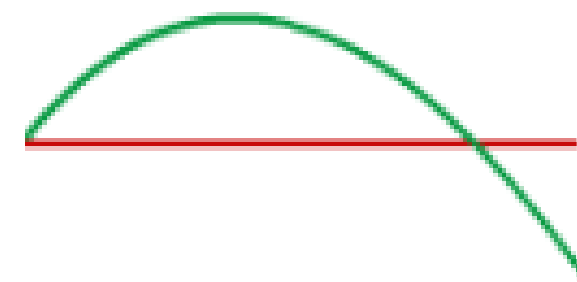




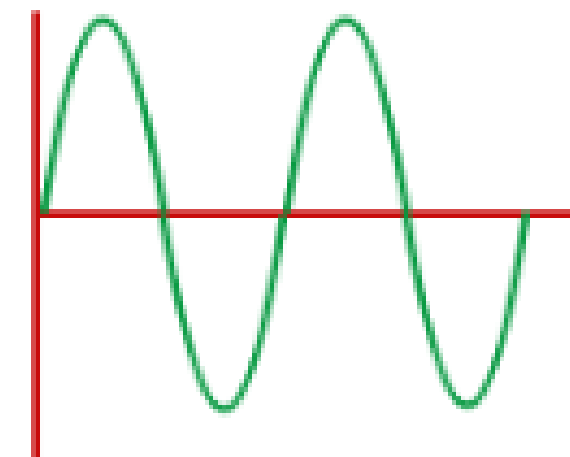
Quieter



Louder



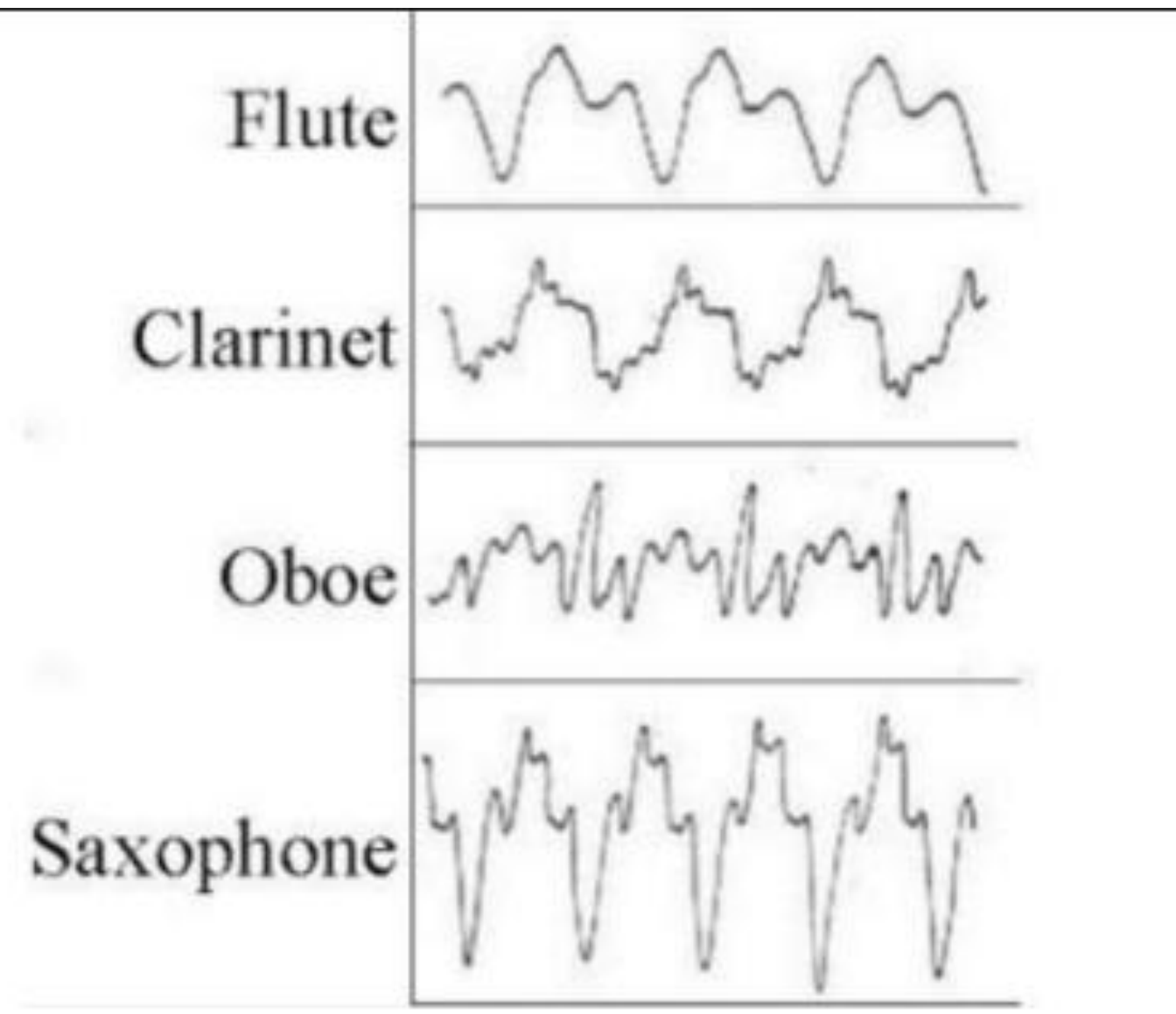
Lower Pitch



Higher Pitch

# Quality or Timbre

- Quality helps us distinguish sounds from different sources (e.g., flute vs guitar).
- Depends on **waveform or harmonics**.
- Same note from different instruments sounds different.



# Noise – A Health Hazard

- **Noise:** Unpleasant, harsh sound made of **irregular vibrations.**
- **Causes:** Motors, vehicles, loudspeakers, etc.

# Effects of Noise Pollution

- Fatigue
- High blood pressure
- Hearing loss (prolonged exposure)
- Emotional and behavioral disturbance

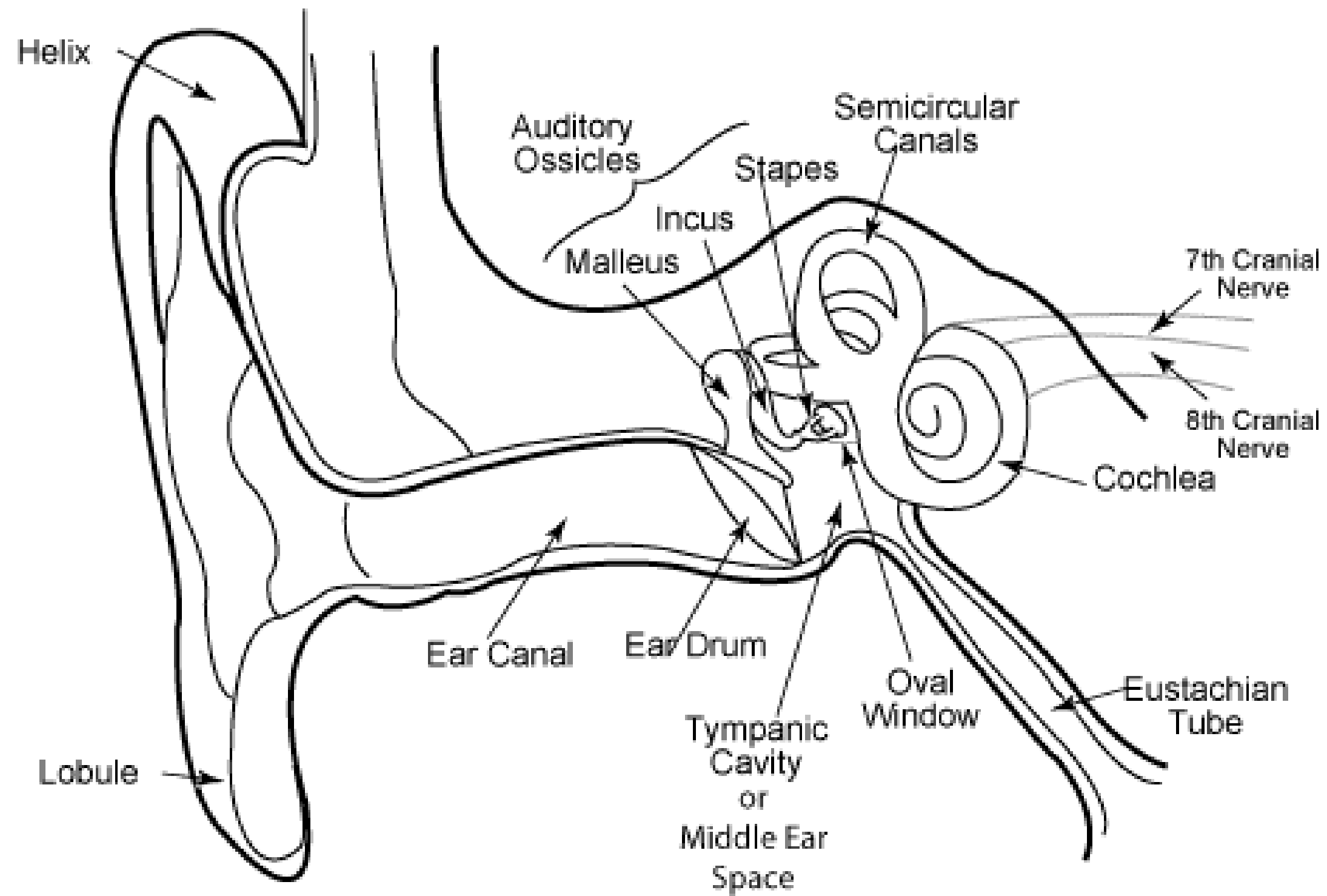


# Ways to Reduce Noise Level

- Move industries away from residential areas
  - Plant trees
- Restrict use of loudspeakers, amplifiers, firecrackers
- Use soft curtains, carpets, cork boards to **absorb sound**

# The Human Ear

- Sound waves enter **outer ear**, reach **eardrum**.
- Vibrations pass to:
  - 1. Hammer**
  - 2. Anvil**
  - 3. Stirrup**
- These bones **amplify** the sound.
- Vibrations pass to the **oval window** → pushes fluid in **cochlea**.
- Thousands of **auditory nerves** in cochlea send messages to the brain.





# Caring for Your Ears

- Ears are **fragile organs**.
- Tips to keep ears healthy:
- Never insert sharp/pointed objects
- Avoid shouting near someone's ear
- Never hit someone near the ear

# Musical Sound vs Noise

- **Musical Sound:** Pleasant, periodic, regular waveform.
  - **Noise:** Harsh, irregular, non-periodic vibrations.
- Instruments create musical sounds due to **harmonics**.

# Instrument Sound Quality

- Instrument shape, size, and structure affect quality of sound.
- Quality helps distinguish different instruments even if they play the same note.

# Formulas Summary

## 1. Speed of Sound:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

## 2. Echo (for depth or distance):

$$\text{Depth} = \frac{1}{2} \times \text{Speed of sound} \times \text{Time}$$

## 3. Frequency & Time Period:

$$\text{Frequency (v)} = \frac{1}{\text{Time Period (T)}}$$

$$\text{Time Period (T)} = \frac{1}{\text{Frequency (v)}}$$

# Important

## **1.Loudness:**

1. Measured in **decibels (dB)**
2. Depends on **amplitude**

## **2.Pitch:**

1. Depends on **frequency**
2. Higher frequency = higher pitch



# Instrument Sound Quality

- Instrument shape, size, and structure affect quality of sound.
- Quality helps distinguish different instruments even if they play the same note.

# Question

- The time period of a pendulum is 0.2 s. What is its frequency?

# Sound vs Light

| Property      | Sound           | Light                     |
|---------------|-----------------|---------------------------|
| Medium needed | Yes             | No (can travel in vacuum) |
| Speed         | ~344 m/s in air | $\sim 3 \times 10^8$ m/s  |
| Type          | Mechanical wave | Electromagnetic wave      |



# KEY POINTS

- ✓ Sound needs a **medium** to travel
- ✓ Human ear hears between **20 Hz to 20,000 Hz**
- ✓ **Ultrasound > 20,000 Hz, Infrasound < 20 Hz**
  - ✓ Echo requires minimum **0.1 second**
- ✓ **Loudness  $\propto$  Amplitude, Pitch  $\propto$  Frequency**

# Recap

- Sound is a **mechanical wave** caused by **vibrations**
- It **reflects, refracts**, and requires a **medium**
- Characteristics: **Loudness, Pitch, Quality**
- Hazards of **noise** and care for the **ears**
- **Ultrasound** and its applications (SONAR, medicine)
- Know your **formulas and units** well for Olympiads

# THANKS!